

S Madhan Kumar

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PROFESSIONAL SUMMARY

A highly skilled Data Analyst and Lead Developer with a Bachelor's in Computer Science specializing in Data Science. Proficient in developing end-to-end analytical pipelines, optimizing workflows, and synthesizing critical business intelligence. Experienced in leveraging Python, Java, SQL, and various AI/ML frameworks to build scalable solutions, including a patent-pending framework for environmental analytics and AI-powered character recognition systems. Demonstrates expertise in data visualization, predictive modeling, and full-stack development to drive strategic initiatives.

TECHNICAL SKILLS

Python

Java

Spring Boot

SQL

C

HTML

JavaScript

TensorFlow

Keras

FastAPI

Scikit-learn

CNN Architecture

Data Pipelines

Git

LM Studio

OpenAI

Aider

Unity

Streamlit

Bootstrap 5

MySQL

Google Earth Engine

Sentinel-2 Analysis

ESP32 Telemetry

SOFT SKILLS

Leadership

Strategic Problem Solving

Communication

EXPERIENCE

Data Science Intern

Jan 2025 - Feb 2025

Ediglobe

- Engineered end-to-end analytical pipelines to solve complex business bottlenecks, contributing to quarterly strategic goal achievement.
- Optimized operational workflows by leveraging data insights, resulting in measurable enhancements to business value and process efficiency.
- Synthesized critical business intelligence for high-level stakeholders, facilitating data-driven decision-making through advanced statistical modeling.
- Developed interactive data visualizations and predictive models to interpret trends and influence strategic company-wide initiatives.

PROJECTS

Multi-Modal MRV Framework

github.com/Madhan1012/Veldora

Developed a patent-pending hybrid framework for Soil Organic Carbon (SOC) estimation using IoT and Satellite data. This system fuses Sentinel-2 multi-spectral imagery with real-time ground telemetry from ESP32 sensors, along with automated data ingestion pipelines for high-precision environmental analytics.

IoT

Satellite Data

Sentinel-2

ESP32

Data Ingestion Pipelines

Ancient Tamil Character Recognition

github.com/Madhan1012/Character-Recognition-V2

Built an AI-powered web application using TensorFlow/Keras and FastAPI to decipher 8th-century Vatteluthu inscriptions. This involved developing a CNN-based OCR model to translate ancient scripts into modern Tamil, preserving historical linguistic data, and featuring a Streamlit frontend.

TensorFlow

Keras

FastAPI

CNN

OCR

Streamlit

NL-to-SQL Intent Engine

github.com/Madhan1012/Natural-Language-To-SQL

Created a modular bot designed to translate natural language queries into executable SQL commands against relational datasets. This engine implemented an intent-based classification system using Scikit-learn to reduce data access friction for non-technical users.

for non-technical users.

SQL Scikit-learn

AegisLM

github.com/Madhan1012/AegisLM

Architected a Template-LM hybrid system utilizing Java Spring Boot and Python to streamline local LLM orchestration. Engineered a complex relational database schema in MySQL with recursive joins and designed integration protocols for local inference servers, including LM Studio and Qwen 2.5 Coder.

Java Spring Boot Python LLM Orchestration MySQL Recursive Joins LM Studio Qwen 2.5 Coder

EDUCATION

Bachelor's of Technology in ComputerScience (Data Science)

2023 - Present

Vel Tech University

12th in MPCB

2022 - 2023

Everwin Vidhyasharam